DHRUV GUPTA

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WORK EXPERIENCE

KYC HUB **Remote**

AI Engineer **Jan 2026 - June 2026**

- Built AI agent workflows to automate onboarding of global AML data sources (sanctions, criminal records, PEPs).
- Implemented an investigative agent to analyze website DOM structures, craft targeted extraction action plans, define attribute schemas, and establish validation logic.
- Engineered a datasource coder agent in Python to generate scrapers, execute web scraping tasks, validate data, and populate downstream processing pipelines.
- Orchestrated extraction and ingestion pipelines with Temporal for durable execution, converting web data into structured compliance records.

SHORTHILLS AI Gurugram, Haryana

Software Developer Intern **Feb 2025 - Aug 2025**

- Built a candidate job matching system using OpenAI embeddings and Elasticsearch retrieval to rank candidates against job requirements, with a recruiter dashboard to review profiles.
- Developed a multi format resume parsing pipeline (LlamaParse, Tesseract OCR) to extract structured data from PDF, DOCX, and scanned documents.
- Automated resume formatting to standardized corporate layouts while rewriting project summaries for clarity.
- Engineered ingestion workflows using Apache NiFi, integrating functionality into Microsoft Teams chat for user access.

HUGHES COMMUNICATION INDIA Gurugram, Haryana

Software Development Intern June 2024 - Aug 2024

- Built a real-time Python terminal tracking system to monitor active network endpoints across distributed locations.
- Integrated geospatial REST APIs with SQL spatial queries and OpenStreetMap to automate geofenced access restrictions.
- Developed a CNN/YOLOv8 model to verify dish antenna placement from field imagery, automating quality sign off and cutting manual inspection time.

PROJECTS

AERO | Agentic Execution for Robotic Operations | GitHub | Live

Tech Stack: Python, JavaScript, Three.js, 3D Simulation, Spatial Mapping

- Engineered a robotics simulation and agentic workflow enabling systems to interpret natural language instructions, plan and execute operations, and ensure spatial reasoning.
- Implemented a simulated 5 DOF robotic arm with geometric inverse kinematics, supporting block stacking and pyramid assembly.

- Designed a deterministic, local execution pipeline that runs without a cloud AI API at runtime, enabling repeatable experiments.
- Developed spatial mapping for object localization, using Three.js to visualize arm motion and execution telemetry.

PaperScope AI | Full-Stack AI Research Assistant | GitHub | Live

Tech Stack: Python, FastAPI, Next.js, React, Google Gemini, Docker

- Built a full stack RAG platform for academic research with PDF analysis, source grounded Q&A, and cross paper synthesis, with page level citations.
- Engineered a configurable ingestion and retrieval pipeline with page aware extraction, user selectable chunking strategies, adjustable top k parameters, and Gemini for embeddings with a TF-IDF fallback for resilience.
- Created a Next.js interface for summaries, methodology and benchmark analysis, BibTeX generation, and Markdown report export.

EDUCATION

DELHI TECHNOLOGICAL UNIVERSITY Dec 2021 - June 2025

Bachelor of Technology (BTech)

Cumulative GPA: 8.13

DOON PUBLIC SCHOOL June 2021

CBSE 12th-89% 10th-88%

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL

Artificial Intelligence: Agentic AI, ML, DL, RAG, NLP, VLA, JEPA

Vision & Robotics: OpenCV, YOLOv8, ROS2, Gazebo

Backends, Distributed Systems & Frameworks: FastAPI, Flask, Django, REST APIs, Temporal, Next.js, React, Streamlit, Three.js, Selenium, Playwright, Pandas, NumPy

Databases, Vector Stores & Search: MongoDB, Redis, FAISS, Elasticsearch, Pinecone

Cloud, DevOps & Environments: Microsoft Azure, Amazon AWS, Google GCP, Docker, Git, Linux

OTHER INFORMATION

- **Codeforces:** Max rating **1508 (Specialist)**, handle: incheif
- **1st Place, Solana Track, LNMHacks 5.0:** Part of a team we built **P2P SOL Marketplace**, a Solana Pay based marketplace for businesses; developed the web interface. Devfolio
- **Co Head, Indian Game Theory Society (DTU):** Led the society and organized events at other colleges, including **Hindu College** and **IIT BHU**.